

StayStill: a large-scale 3D idle animation dataset

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Abstract

Motivation

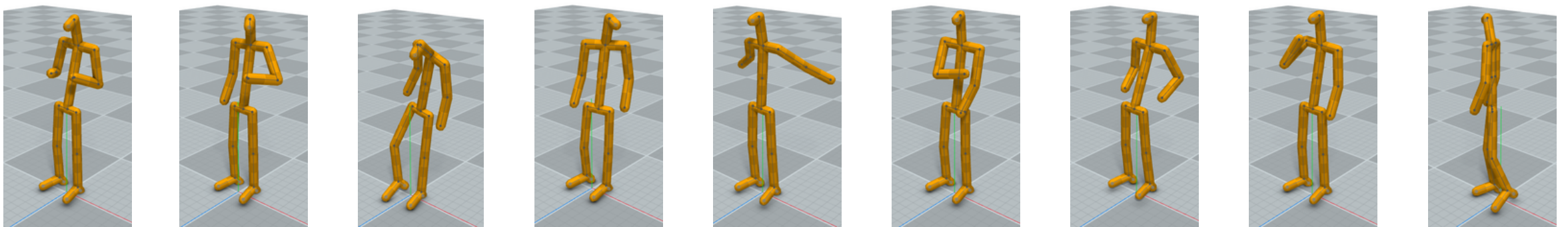
- **Idle animations** are essential for character realism.
- Researching idle motion and its unique dynamics can lead to very interesting **idle animation generation** research.
- However, **currently**, there are **no large public 3D idle animation datasets** available.
- Research on idle motion generation is very scarce.
- This work in progress dataset is the **first publicly available large-scale 3D idle animation dataset**. It is constructed with data-hungry deep-learning applications in mind.

Contributions

- **Construct** a large **dataset** specifically for idle motion
- **Validate** the dataset using deep learning, to **ensure** that it is **useful to train deep learning** networks
- **bvhTools**: an open-source python library to work with BVH files

Dataset Details

- **50 subjects**
- **~5 hours** of idle motion data (~500.000 frames)
 - **General** idle (1h 45min)
 - Idle **with a phone** (1h 45min)
 - Idle **actions** (3 hours)
- 18 different **idle action classes**:
 - Look around, look back(left/right), look down, check phone, look at shoes, look up/sky, check watch, scratch (arm, back, leg, head, eyes), stretch (arms, back), touch face, change balance, yawn
- BioVision Hierarchy (**BVH**) format
- **2 skeleton** structures (Freemocap, LaFAN1)



Recording procedure

Used a multi-camera recording setting, alongside the **Freemocap software** (**no** need for a motion capture **suit**). This way, we achieve a **less intrusive** recording procedure.

We recorded **non-actors** on an **empty room** setting, to avoid distractions and interaction with the environment.

3 recording phases:

- We asked them to perform 18 **actions**, twice
- We asked them to **stand** not doing anything specific
- We asked them to repeat the process, but **while using a phone**

Data validation

The data is **currently being validated**, to check whether deep-learning based methods are able to learn specific motion dynamics present on idle animations.

Task: motion in-betweening

We are currently **testing the data on state-of-the-art motion in-betweening networks**, and **comparing** them to 2 typical **baselines**:

- **SLERP**
- **0-velocity**

bvhTools library

We also developed a **python library** to process and **work with** the BVH (BioVision Hierarchy) **animation files**. It is extremely simple to use, useful for typical machine learning pipelines. It is also **open source**.

It is already usable: check the **repository** and **documentation** here!

<https://enekoassets.github.io/bvhTools/#/>



Check the repository

The repository does **not currently contain the data**, but you can give it a **star** to follow the development of the dataset.

Planned release: after validating the data and publishing the paper on arXiv.

<https://github.com/Enekoassets/StayStill>



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